

Question	Answer	Marks
8(a)(i)	half-wave (rectification)	B1
8(a)(ii)	$V_0 = 6.0 \times \sqrt{2}$ $= 8.5 \text{ V}$	A1
8(b)(i)	$P = V^2 / R$	C1
	$P_0 = 8.5^2 / 45$ $= 1.6 \text{ W}$	A1
8(b)(ii)	two humps of width $0.5T$ and two sections of zero power of width $0.5T$	B1
	all humps drawn have width $0.5T$, minima at $P = 0$ and peaks at $P = P_0$	B1
	correct sinusoidal shape, with smooth troughs sitting on t -axis at $P = 0$	B1
8(b)(iii)	mean power within each hump is $\frac{1}{2}P_0$ from the symmetry of the curve	B1
	additional half factor from removal of half of the power in each cycle	B1
8(b)(iv)	$\langle P \rangle = V_{\text{r.m.s.}}^2 / R$ $V_{\text{r.m.s.}} = \sqrt{[(1.6 / 4) \times 45]}$ $= 4.2 \text{ V}$	A1

9702/42

Cambridge International AS & A Level – Mark Scheme
PUBLISHED

May/June 2025

Question	Answer	Marks